

Notes: Liu and Pan (WP, 2025)

Pricing the Global Trade Vulnerability

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1 Big picture

- **Question.** Has global trade vulnerability become a priced systematic risk in U.S. equity markets?
- **Core idea.** A firm that imports from only a few countries is more exposed to trade-policy shocks, geopolitical shocks, tariffs, and supply-chain disruptions than a firm with diversified sourcing. The paper asks whether the stock market prices that exposure.
- **Data innovation.** The paper uses granular S&P Global Panjiva bill-of-lading data to observe firm-level imports by country, supplier, and product. This lets the authors build firm-level measures of trade-network concentration rather than relying on industry-level import exposure.
- **Main measure.** Trade vulnerability is measured by the Herfindahl concentration of a firm's import value across source countries, denoted country concentration, or CC.
- **Portfolio design.** Each month, importing firms are sorted into low, medium, and high country-concentration portfolios ($P1, P2, P3$). Firms with no import records form the zero-trade benchmark $P0$. The main factor is

$$R_t^{Trade} = R_t^{P3} - R_t^{P0}.$$

- **Main result.** The high-minus-zero trade factor $HL30 = P3 - P0$ earns about 0.87% per month in average excess return. It remains significant after CAPM and Fama–French five-factor adjustment, with a five-factor alpha of about 0.55% per month.
- **Economic mechanism.** The premium is not simply import intensity. It is associated with concentrated sourcing networks. Firms in $P3$ import from fewer countries, fewer suppliers, and fewer product categories than firms in $P1$.
- **China result.** China exposure amplifies the premium only when combined with high country concentration. China dependence by itself is not the central priced object; concentrated China dependence is.
- **Industry and technology result.** High country-concentration firms are disproportionately in technology-intensive industries, especially business equipment and semiconductors. But industry-neutral and within-industry tests show that the premium survives after removing industry tilts.
- **Dynamic result.** Volatility spreads and event studies around the 2018–2019 U.S.–China trade war, COVID-19, and the 2025 tariff episodes show that trade vulnerability becomes more important precisely when trade-policy risk becomes more salient.
- **Economic takeaway.** Global trade exposure is not just a real-side supply-chain issue. In this paper, concentrated global sourcing becomes an asset-pricing risk: investors require compensation for holding firms whose production networks are fragile to trade shocks.

2 Data and measurement (key objects)

2.1 Shipment-level import data

The paper’s core dataset is S&P Global Panjiva, which is built from U.S. Customs and shipping-company bills of lading. Each shipment-level record contains information such as the U.S. consignee, foreign shipper, country of origin, shipment date, HS product code, weight, container counts, and declared import value.

The broad sample covers U.S. import transactions from 2012 to 2023. After cleaning, the full Panjiva import sample contains more than 102 million shipment observations.

Interpretation. The measurement contribution is important. Instead of saying that a whole industry is exposed to trade risk, the paper can say that a specific public firm imports heavily from a specific country, supplier, or product category.

2.2 Cleaning and public-firm matching

The authors remove transactions with missing import value, weight, origin, destination region, or HS code. They also restrict to shipments in which all goods in the transaction share the same two-digit HS product code, so that the import value can be cleanly assigned to a product category.

For public firms, the paper matches Panjiva consignee identifiers to Compustat using S&P Global Company ID. It then aggregates shipments to the firm-year level and merges the import measures with Compustat financial variables. The public-firm import sample contains about 2.66 million shipment-level observations, 1,840 unique firms, and roughly \$469.9 billion of cumulative import value over 2012–2023.

Interpretation. The match allows the paper to move from trade records to stock returns. Without the Panjiva–Compustat–CRSP bridge, country concentration would be a trade statistic rather than an asset-pricing characteristic.

2.3 Representativeness of the Panjiva data

The paper validates Panjiva against aggregate trade data. For total U.S. imports, Panjiva captures roughly 40–46% of BEA import values over 2012–2023 and moves closely with the aggregate series. For imports from China, Panjiva captures roughly 22–45% of UN Comtrade import values and tracks the same broad dynamics.

Interpretation. Panjiva is not the universe of all imports, especially because bill-of-lading data emphasize marine shipments. But the high time-series comovement with official aggregate sources supports using Panjiva to measure firm-level variation in import-network exposure.

2.4 Country concentration

For firm i , country c , and shipment year t , let $I_{i,c,t}$ be the import value from country c . Total imports are

$$I_{i,t} = \sum_c I_{i,c,t}. \quad (1)$$

The import share from country c is

$$s_{i,c,t} = \frac{I_{i,c,t}}{I_{i,t}}. \quad (2)$$

The country concentration index is

$$CC_{i,t} = \sum_c s_{i,c,t}^2 = \sum_c \left(\frac{I_{i,c,t}}{I_{i,t}} \right)^2. \quad (3)$$

Interpretation. This is a Herfindahl index. A firm importing everything from one country has $CC = 1$. A firm importing evenly from N countries has $CC = 1/N$. Higher CC means greater dependence on a narrow set of countries and therefore greater vulnerability to country-specific trade disruptions.

2.5 Industry-demeaned country concentration

Because import patterns differ systematically across industries, the baseline sorting variable is industry-demeaned country concentration:

$$CC_{i,t}^{demean} = CC_{i,t} - \overline{CC}_{I(i),t}, \quad (4)$$

where $I(i)$ is firm i 's Fama–French 48 industry and $\overline{CC}_{I(i),t}$ is that industry's average CC in year t .

Interpretation. This asks whether a firm is unusually concentrated relative to other firms in its industry. That is useful because some industries are naturally more global or more China-dependent than others.

2.6 Portfolio objects

At each month-end, stocks are sorted into:

- $P0$: firms with no Panjiva import records in the relevant shipment year;
- $P1$: low country concentration among importers;
- $P2$: medium country concentration among importers;
- $P3$: high country concentration among importers.

The paper uses NYSE breakpoints, value-weighted portfolio returns, a price screen excluding stocks below \$5, and delisting-bias corrections. To avoid look-ahead bias, portfolios formed before June use trade data from two years earlier, while portfolios formed in June or later can use the previous year's shipment data.

Interpretation. The timing convention is conservative. It ensures that the portfolio sort uses only import information that would plausibly have been available when the portfolio was formed.

2.7 Trade-risk factors

The main trade-risk factor is

$$R_t^{Trade} = R_t^{P3} - R_t^{P0}. \quad (5)$$

The paper calls this high-minus-zero factor HL30. It also studies

$$\text{HL31}_t = R_t^{P3} - R_t^{P1}, \quad (6)$$

which compares high-concentration importers with diversified importers.

Interpretation. HL30 asks whether concentrated import exposure is rewarded relative to no direct trade exposure. HL31 asks whether concentrated import exposure is rewarded relative to diversified import exposure. The first is the preferred benchmark because $P0$ is the cleanest zero-trade comparison group.

2.8 Alternative vulnerability measures

The paper constructs several alternative measures:

- supplier concentration (SC), based on import shares across foreign suppliers;
- goods concentration (GC), based on import shares across HS2 product categories;
- inverse counts of countries, suppliers, and goods ($1/NC, 1/NS, 1/NG$);
- import-to-revenue ratio;
- China import ratio.

Interpretation. These measures separate two ideas. One idea is *how much* a firm imports. Another idea is *how concentrated* its sourcing is. The paper’s main result is about concentration, not just the level of imports.

3 Models and empirical specifications (all equations + interpretation)

3.1 Factor-pricing logic

The paper’s economic hypothesis is that global trade vulnerability is a systematic state variable. When trade becomes risky, firms with concentrated sourcing should underperform diversified or zero-trade firms. If that bad state is also a bad state for investors, the high-vulnerability portfolio must offer a positive expected return as compensation.

In a two-factor pricing model, the paper writes the excess return of asset i as

$$R_{i,t} - r_{f,t} = \alpha_i + \beta_i^M (R_t^{Mkt} - r_{f,t}) + \beta_i^T R_t^{Trade} + \varepsilon_{i,t}. \quad (7)$$

Interpretation. The trade beta β_i^T measures exposure to the trade-vulnerability factor. If the factor is priced and correctly included, assets should not have abnormal returns after controlling for market and trade exposure.

3.2 Main time-series pricing regression

For each portfolio $T \in \{P0, P1, P2, P3, \text{HL30}, \text{HL31}\}$, the paper estimates

$$R_t^T = \alpha_T + \beta_T' \mathbf{f}_t + \varepsilon_t^T, \quad (8)$$

where $\mathbf{f}_t$ is either the market excess return or the Fama–French five factors.

Interpretation. The intercept α_T is the risk-adjusted return. A positive and significant α for $P3$, HL30 , or HL31 means that standard asset-pricing factors do not fully explain the return premium associated with concentrated sourcing.

3.3 The main pricing results

The main monthly portfolio results are:

$$E[R^{P0}] \approx 0.84\%, \quad E[R^{P1}] \approx 0.80\%, \quad (9)$$

$$E[R^{P2}] \approx 1.04\%, \quad E[R^{P3}] \approx 1.71\%, \quad (10)$$

$$E[\text{HL30}] \approx 0.87\%, \quad E[\text{HL31}] \approx 0.91\%. \quad (11)$$

Under the Fama–French five-factor model, $P3$ still earns an alpha of about 0.52% per month, HL30 earns about 0.55%, and HL31 earns about 0.64%.

Interpretation. These magnitudes are large in monthly terms. The result is not just that high-concentration firms have higher raw returns; the spread survives standard adjustments for market, size, value, profitability, and investment factors.

3.4 Alternative measures and spanning tests

For alternative trade measures X , the paper forms analogous $P3 - P0$ long-short portfolios. It then asks whether their alphas survive once the CC-based trade factor is included:

$$R_t^{X, P3-P0} = \alpha_X + \beta_X^M (R_t^{Mkt} - r_{f,t}) + \beta_X^T R_t^{Trade} + \varepsilon_{X,t}. \quad (12)$$

Interpretation. This is a spanning test. If supplier concentration, goods concentration, or China ratio have alphas under CAPM but not after adding R^{Trade} , then their pricing information is largely contained in the country-concentration factor.

Empirically, supplier concentration, goods concentration, and inverse number of countries have positive Fama–French alphas, but their alphas largely disappear once the CC-based trade factor is included.

3.5 China double sort

The paper then asks whether the main premium is really about China. Within each CC portfolio, firms are split into high- and low-China-import-ratio groups. The key comparison is a difference-in-differences:

$$(R_{HighChina}^{P3} - R_{HighChina}^{P1}) - (R_{LowChina}^{P3} - R_{LowChina}^{P1}). \quad (13)$$

Interpretation. This design asks whether China exposure matters more when a firm is also country-concentrated. The answer is yes. High country concentration plus high China exposure earns the strongest premium, while China exposure alone is not a complete substitute for the country-concentration measure.

3.6 Industry-neutral portfolios

To test whether the result is an industry tilt, the paper constructs industry-neutral returns. For each CC portfolio p and industry j , it first computes a value-weighted return $R_{p,j,t+1}$. It then reweights industries using benchmark industry weights $\hat{\omega}_{j,t}$:

$$R_{p,t+1}^{INT} = \sum_{j=1}^{n_p} \hat{\omega}_{j,t} R_{p,j,t+1}. \quad (14)$$

For long-short portfolios, the paper restricts the two legs to the intersection of industries appearing in both legs, so the long and short sides share the same industry composition.

Interpretation. This removes the concern that $P3$ earns high returns merely because it overweights technology or semiconductor firms. The premium falls after industry neutralization, but remains positive and statistically significant.

3.7 Within-industry sorting

As a stricter test, the paper sorts firms into CC portfolios separately within Fama–French 12 industries and then aggregates across industries.

Interpretation. This removes cross-industry variation by construction. Any remaining premium comes from firms that are more country-concentrated than other firms in the same broad industry.

The within-industry results are strong: $P3$ earns about 1.70% per month and HL30 earns about 0.67% per month in excess returns. The five-factor alpha of HL30 remains about 0.56% per month.

3.8 Technology exposure regressions

Because high-CC firms are often in technology-intensive industries, the paper tests whether technology indices can explain the trade factor. It estimates

$$R_t^i = \alpha_i + \beta_i^M (R_t^{Mkt} - r_{f,t}) + \beta_i^{TC} R_t^{Tech} + \varepsilon_{i,t}, \quad (15)$$

where R_t^{Tech} is a broad technology index or a semiconductor index.

Interpretation. If the trade factor were just a disguised technology factor, technology controls should eliminate its alpha. They do not. The trade factor remains positively priced after controlling for broad tech and semiconductor proxies.

The reverse test prices technology and semiconductor portfolios using market plus trade:

$$R_t^{Tech} = \alpha + \beta^M (R_t^{Mkt} - r_{f,t}) + \beta^T R_t^{Trade} + \varepsilon_t. \quad (16)$$

Interpretation. This is where the paper makes a stronger economic point: some high technology and semiconductor outperformance looks like compensation for exposure to global supply-chain fragility, not only pure technology-sector risk.

3.9 Volatility-spread analysis

For each vulnerable portfolio $j \in \{P1, P2, P3\}$, the paper estimates time-varying volatility and compares it with the zero-trade portfolio:

$$\Delta_{j0,t} = \sigma_{j,t} - \sigma_{0,t}. \quad (17)$$

The volatilities are estimated using GMM moment conditions with exponentially decaying weights.

Interpretation. The return premium is the average compensation. The volatility spread shows when the market perceives trade vulnerability to be especially risky. Spikes around trade-war, COVID, and tariff episodes support the idea that the factor is connected to real trade uncertainty.

3.10 Liberation Day event study

The paper estimates daily Fama–French five-factor betas using 2024 data, then calculates cumulative abnormal returns from January 2, 2025 through June 30, 2025.

Interpretation. The pre-2025 betas define the expected return path. Deviations around the tariff events show how the market reprices different vulnerability groups when new trade-policy information arrives.

The key pattern is layered. Broad tariff news on April 2 hurts medium-vulnerability firms ($P2$) the most. The later China-specific escalation hits the most concentrated and China-dependent firms ($P3$) more sharply.

4 Identification and credibility

4.1 What identifies the trade-risk premium?

The paper is not a causal treatment-effects paper. The identifying object is a priced return spread. The core empirical claim is that portfolios sorted on pre-available country-concentration measures earn different future returns, and that this return difference is not explained by standard factors or obvious alternative characteristics.

The main credibility comes from five design choices:

- direct firm-level import measurement from bills of lading;
- conservative data timing to avoid look-ahead bias;
- value-weighted portfolio construction with standard CRSP screens;
- risk adjustment using CAPM and Fama–French five factors;
- industry-neutral and within-industry portfolio tests.

4.2 Why country concentration is not just import intensity

The paper explicitly compares CC with the import-to-revenue ratio. Import intensity does not generate the same pricing pattern. In Table 4, portfolios sorted on import-to-revenue do not show a positive and significant high-minus-zero trade premium.

Interpretation. The market is not simply rewarding firms that import a lot. It is rewarding exposure to a fragile sourcing structure: dependence on a small number of countries.

4.3 Why country concentration is not only a China factor

The China import ratio by itself does not produce the same robust premium as country concentration. The strongest evidence appears when China dependence is interacted with high concentration.

Interpretation. This distinction is important. China is a central node in the empirical premium, especially after the 2018 trade war and COVID disruptions, but the deeper mechanism is concentrated trade exposure. China is a powerful amplifier of that mechanism.

4.4 Why the result is not merely an industry or technology effect

The paper uses two industry controls. First, global industry-neutral weights make portfolios comparable in industry composition. Second, within-industry sorting compares firms with different trade concentration inside the same broad industry.

The premium survives both tests. It is smaller in the industry-neutral tests, meaning industry composition matters, but it does not disappear.

Interpretation. The correct interpretation is not “this is unrelated to technology.” Technology sectors are economically central because their supply chains are global and concentrated. The correct interpretation is that trade vulnerability is not mechanically identical to being a technology stock.

4.5 Why the result is economically timed to trade-risk episodes

The trade factor’s cumulative returns and volatility spreads move around major trade and supply-chain events: the 2018–2019 U.S.–China trade war, COVID-19 supply-chain disruptions, the 2024 election, and the 2025 tariff episodes.

Interpretation. Timing evidence does not prove causality by itself, but it strengthens the economic story. The factor behaves like a trade-risk factor when trade risk becomes salient.

4.6 Remaining limitations

- Panjiva does not cover every mode of import, so some firm exposures are imperfectly measured.
- The sample period is short for asset-pricing tests, especially because the strongest trade-risk events cluster after 2018.

- The zero-trade portfolio may contain firms with indirect global exposure that is not visible in import records.
- The paper prices concentrated sourcing, but it does not fully disentangle all real channels: tariffs, supplier disruption, geopolitical risk, inventory management, and demand-side effects can interact.

Interpretation. These limitations do not overturn the evidence, but they define the scope of the claim. The paper documents a robust return pattern associated with measured trade concentration; it does not claim to observe every dimension of a firm’s global supply-chain exposure.

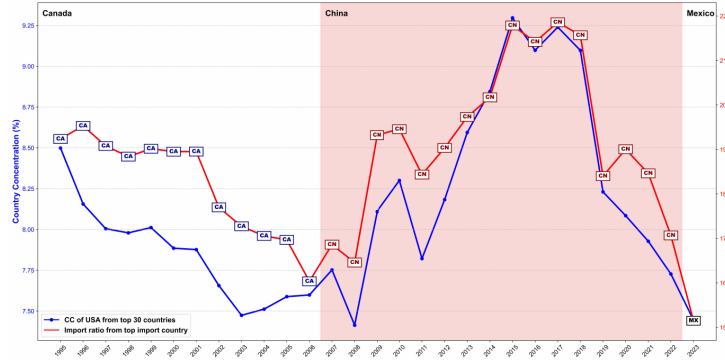
5 Tables and figures

5.1 Figure 1: U.S. import concentration and top-country share

Read: The figure plots aggregate U.S. import country concentration and the share of imports from the largest partner. China becomes the dominant import partner during the 2000s and 2010s, and aggregate concentration rises alongside China’s import share.

Economic message. The macro environment motivates the firm-level measure. If the country-level U.S. import structure becomes more concentrated, firm-level concentration can become a source of systematic rather than purely idiosyncratic risk.

Figure 1: Top Country Import Ratio vs. Import Country Concentration for the U.S.



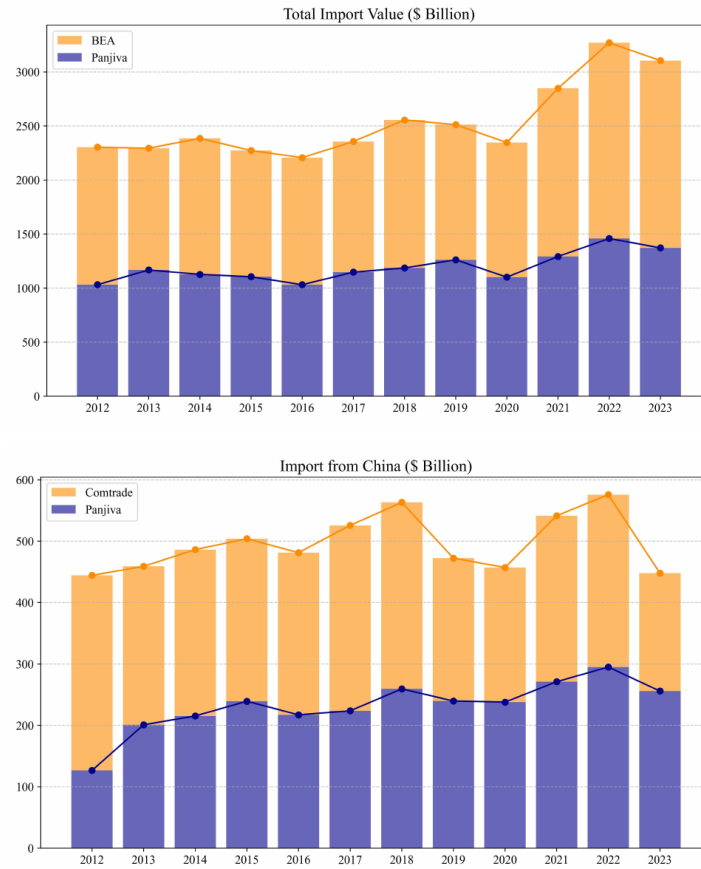
Aggregate import concentration and the top-country import share.

5.2 Figure 2: Panjiva versus official aggregate import data

Read: The figure compares Panjiva import values with BEA total imports and UN Comtrade imports from China.

Economic message. Panjiva is incomplete but representative in dynamics. This supports using it for cross-sectional firm-level exposure measures.

Figure 2: U.S. Import Summary from S&P Panjiva, BEA, and Comtrade



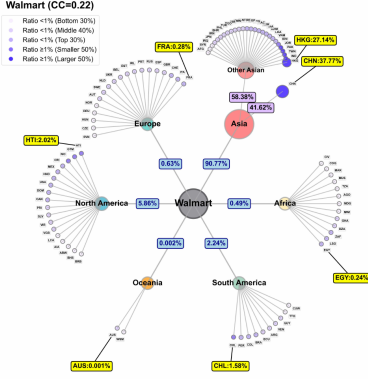
Panjiva compared with BEA and Comtrade import series.

5.3 Figure 3: Walmart, Tesla, and Apple trade networks

Read: The three examples show a spectrum: Walmart has low concentration ($CC \approx 0.22$), Tesla is intermediate ($CC \approx 0.39$), and Apple is highly concentrated ($CC \approx 0.88$), with very large exposure to China.

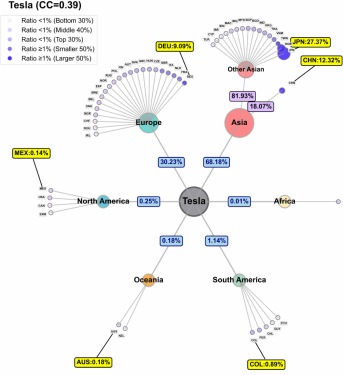
Economic message. These examples make the measure intuitive. Two firms can both be global importers, but one can be much more fragile if its import network is concentrated in one country.

Figure 3(a): Walmart Global Trade Network



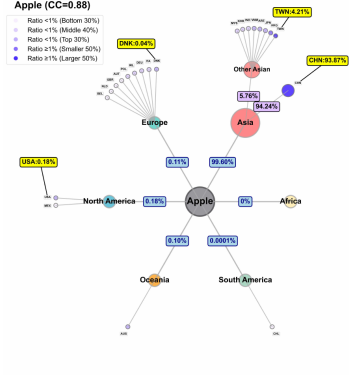
Walmart

Figure 3(b): Tesla Global Trade Network



Tesla

Figure 3(c): Apple Global Trade Network



Apple

Examples of low, medium, and high trade-network concentration.

5.4 Table 1: Summary statistics

Read: The average public-firm importer is highly skewed: mean annual import value is about \$44 million, while the median is much smaller. Firms import from about 5–6 countries on average, with Asia and China central among positive importers.

Economic message. The firm-level import network is very heterogeneous. This heterogeneity is necessary for the asset-pricing design: some firms are diversified, while others are strongly dependent on a narrow set of countries.

Table 1: Summary Statistics All Sample

This table presents summary statistics for the bill-of-lading data used in our analysis. For each shipment year from 2012 to 2023, we compute the time-series average of year-specific cross-sectional statistics. Import Value denotes the total value of a firm's imports in a given year, aggregated across all shipments. The Import/Revenue ratio is defined as the firm's total import value divided by its total annual revenue. The counts of import countries, suppliers, and HS goods capture the number of distinct origin countries, trading partners, and HS product codes from which a firm imports. The Import Ratio by Location measures the fraction of a firm's total import value sourced from a given geographic area. This statistic is computed conditionally: the ratio is reported only for firm-area pairs with strictly positive import flows.

Trade Variable	Mean	Std	P1	P25	Median	P75	P99
Import value (in thousands)	44,390	247,911	0.85	151	1,567	11,976	723,289
Import/Revenue ratio (%)	2.25	14.41	0.00	0.01	0.11	0.81	34.68
Import from number of							
Countries	5.62	6.86	1.00	1.15	3.00	6.92	35.33
Suppliers	19.94	60.91	1.00	1.92	4.92	15.71	229.80
HS2 Goods	5.94	7.20	1.00	1.25	3.17	7.42	35.40
Import ratio by location (%)							
Asia	75.05	34.26	0.20	55.06	95.89	100	100
China	42.03	36.84	0.03	7.11	31.44	78.17	100
Europe	43.58	40.07	0.02	4.44	30.66	90.52	100
North America	22.88	34.49	0.00	0.40	3.33	31.42	100
South America	17.42	29.35	0.00	0.32	2.63	18.98	100
Africa	12.20	25.30	0.00	0.20	1.30	8.12	97.81
Oceania	14.18	28.61	0.00	0.10	0.95	8.56	100

Summary statistics of firm-level import measures.

5.5 Table 2: Characteristics of CC-sorted portfolios

Read: $P3$ firms have much higher country concentration, supplier concentration, and goods concentration than $P1$ firms. They import from fewer countries, fewer suppliers, and fewer HS2 goods categories.

Economic message. The portfolio sort does what it should do. The high-vulnerability portfolio is genuinely more concentrated in trade-network structure, not merely relabeled by another arbitrary characteristic.

Table 2: Summary Statistics of the CC-Sorted Portfolios

This table reports summary statistics for the trade measures across portfolios sorted on firms' country concentration. Country Concentration (CC) is computed following equation (1). Industry-demeaned Country Concentration is constructed using equation (2). Supplier Concentration and HS2 Goods Concentration are defined analogously to CC, but replace the country dimension with the supplier and product (HS2) dimensions, respectively. Detailed definitions of all measures are provided in the Appendix. Number of Countries, Suppliers, and HS2 Goods denote the counts of distinct origin countries, trading partners, and HS product categories from which a firm imports. Import Value is the total value of a firm's imports in a given year, aggregated across all shipments. The Import/Revenue ratio is the firm's total import value divided by its total annual revenue. Number of Firms reports the number of firms in each portfolio. Market Cap and Revenue represent the average market value and annual revenue of firms within each portfolio. ROA is defined as the ratio of net income to total assets, while Operating Margin is measured as operating income after depreciation divided by sales. All statistics are reported as the time-series average of the cross-sectional mean or median, as indicated.

	Mean				Median			
	P1	P2	P3	P0	P1	P2	P3	P0
Country Concentration (industry demeaned)	-0.31	-0.05	0.28		-0.29	-0.06	0.27	
Country Concentration	0.31	0.55	0.91		0.30	0.53	0.97	
Supplier Concentration	0.22	0.41	0.78		0.21	0.41	0.88	
HS2 Goods Concentration	0.51	0.64	0.83		0.47	0.63	0.95	
Number of Countries	12.20	7.11	2.99		9.24	5.13	2.00	
Number of Suppliers	49.05	28.22	7.58		20.69	10.20	2.90	
Number of HS2 Goods	11.45	7.67	3.80		8.82	5.38	2.13	
Import Value (\$ million)	101	72	28		10	5	1	
Import/Revenue Ratio (%)	1.65	2.00	1.79		0.34	0.24	0.11	
Number of Firms	175	195	229	2190				
Market Cap (\$ million)	23,555	23,076	18,406	6,517	3,909	2,418	1,638	1,074
Revenue (\$ million)	13,910	9,170	6,730	3,887	3,279	2,026	1,197	582
ROA (%)	5.22	4.48	2.38	-2.24	5.45	5.01	4.36	1.51
Operating Margin (%)	10.47	7.95	-22.91	-98.46	9.95	9.35	8.58	11.91

Portfolio characteristics sorted by country concentration.

5.6 Table 3 and Figure 4: Main pricing evidence

Read: Table 3 shows that $P3$ earns much higher returns than $P0$, $P1$, and $P2$. The HL30 factor earns 0.87% per month and remains significant after CAPM and Fama–French adjustment. Figure 4 shows the cumulative factor performance over time.

Economic message. This is the paper's core asset-pricing result. Concentrated global sourcing commands a positive risk premium that standard factors do not absorb.

Table 3: Main Pricing Results of Trade Vulnerability Portfolios by Country Concentration

This table reports the main asset-pricing results for the trade-vulnerability portfolios constructed using firms' import country concentration. The value-weighted portfolios P0 through P3 sort firms from zero trade exposure (P0) to the highest trade vulnerability (P3). The upper panel reports the Pearson correlation matrix between the country-concentration factors (HL30 and HL31) and the Fama–French five factors. HL30 is defined as the high-minus-zero portfolio ($P3 - P0$), and HL31 is defined as the high-minus-low portfolio ($P3 - P1$). The middle panel reports the excess returns of the four trade-vulnerability portfolios and the excess returns of the two long-short factors (HL30 and HL31). The lower panel reports the asset-pricing results for each portfolio and each long-short factor under the CAPM and the Fama–French five-factor model. We report risk-adjusted returns (alphas) and factor loadings (betas) estimated from each model. The sample covers January 2014 to June 2025. Newey–West (1987) t -statistics are shown in square brackets.

Person Correlation Matrix						
	HL31	MKT	SMB	HML	RMW	CMA
HL30	0.84	0.14	-0.29	-0.42	0.26	-0.28
HL31		0.30	-0.03	-0.44	-0.06	-0.45

Pricing Results						
Excess Ret (%)	P0	P1	P2	P3	HL30	HL31
	0.84	0.80	1.04	1.71	0.87	0.91
	[2.78]	[3.02]	[2.83]	[4.51]	[3.76]	[3.49]

CAPM						
α (%)	-0.13	-0.04	0.05	0.65	0.78	0.69
	[-1.83]	[-0.37]	[0.29]	[3.53]	[3.44]	[2.82]
β^{MKT}	1.01	0.87	1.02	1.10	0.09	0.23
	[53.26]	[33.43]	[23.45]	[25.24]	[1.56]	[3.79]

Fama-French Five-Factor						
α (%)	-0.03	-0.12	0.00	0.52	0.55	0.64
	[-0.46]	[-1.12]	[-0.01]	[3.25]	[2.88]	[2.87]
β^{MKT}	0.98	0.89	1.03	1.09	0.11	0.20
	[88.62]	[32.11]	[24.58]	[28.40]	[2.48]	[3.70]
β^{SMB}	0.08	-0.03	-0.06	-0.04	-0.11	-0.01
	[3.13]	[-0.80]	[-1.03]	[-0.48]	[-1.40]	[-0.06]
β^{HML}	0.12	0.10	-0.22	-0.19	-0.31	-0.29
	[4.46]	[1.86]	[-3.64]	[-2.45]	[-3.53]	[-2.42]
β^{RMW}	-0.14	0.22	-0.04	0.26	0.39	0.04
	[-3.99]	[3.26]	[-0.48]	[2.70]	[3.34]	[0.31]
β^{CMA}	-0.09	0.15	0.01	-0.16	-0.06	-0.31
	[-1.88]	[1.58]	[0.08]	[-1.10]	[-0.36]	[-1.47]

Main pricing results for country-concentration portfolios.

Figure 4: Cumulative Returns of Trade Vulnerability Portfolios against Zero Trade Exposure Portfolio

Cumulative returns of trade-vulnerability portfolios.

5.7 Table 4: Alternative trade-vulnerability measures

Read: Supplier concentration, goods concentration, and inverse number of countries carry some pricing information. But their alphas are largely absorbed when the CC-based trade factor is included.

Economic message. Country concentration is the most robust summary statistic for the priced component of trade vulnerability.

Table 4: Alternative Trade Vulnerability Measures and Pricing Results

This table reports the asset-pricing results for trade-vulnerability portfolios constructed using a set of alternative trade measures. Specifically, we consider Supplier Concentration (SC), HS2 Goods Concentration (GC), the inverse of the number of countries (1/NC), the inverse of the number of suppliers (1/NS), and the inverse of the number of HS2 product categories (1/NG). In addition, we examine two value-based measures: the firm's Import/Revenue ratio and its China Ratio, defined as the share of total import value sourced from China in a given shipment year. Panel A reports the risk-adjusted returns (alphas) from the Fama-French five-factor model for the value-weighted portfolios sorted on each alternative trade measure. Panel B compares the CAPM with the CAPM augmented by the country-concentration trade vulnerability factor in pricing the high-minus-zero (P3 – P0) value-weighted portfolios constructed from these alternative measures. The sample covers January 2014 to June 2025. Newey–West (1987) *t*-statistics are shown in square brackets.

Panel A: Fama-French 5-Factor Alpha (%)

Alternative Measures	No Trade	Sorted Portfolios			Long/Short	
	P0	P1	P2	P3	P3 – P0	P3 – P1
Supplier Concentration (SC)	-0.03 [-0.46]	0.05 [0.57]	-0.11 [-0.65]	0.42 [2.31]	0.44 [2.08]	0.36 [1.81]
HS2 Goods Concentration (GC)	-0.03 [-0.46]	0.08 [0.79]	-0.04 [-0.24]	0.33 [2.05]	0.36 [2.01]	0.25 [1.43]
1/Number of Countries (1/NC)	-0.03 [-0.46]	0.19 [1.89]	-0.32 [-2.05]	0.54 [2.18]	0.57 [2.13]	0.36 [1.26]
1/Number of Suppliers (1/NS)	-0.03 [-0.46]	0.26 [2.51]	-0.34 [-3.07]	0.42 [1.81]	0.44 [1.81]	0.16 [0.61]
1/Number of HS2 Goods (1/NG)	-0.03 [-0.46]	0.18 [2.12]	-0.22 [-1.44]	0.22 [0.92]	0.24 [0.95]	0.03 [0.13]
Import/Revenue (Import)	-0.03 [-0.46]	0.24 [1.44]	0.13 [1.30]	-0.15 [-1.02]	-0.12 [-0.72]	-0.39 [-1.51]
China Ratio (China)	-0.03 [-0.46]	0.11 [0.52]	-0.07 [-0.60]	0.22 [1.55]	0.25 [1.41]	0.11 [0.40]

Panel B: Pricing for P3 – P0 of Other Trade Measures

Alternative Measures	SC	GC	1/NC	1/NS	1/NG	Import	China
CAPM							
α (%)	0.54 [2.42]	0.43 [2.20]	0.81 [2.63]	0.58 [2.19]	0.41 [1.41]	0.01 [0.09]	0.52 [2.01]
β_{MKT}	0.06 [0.93]	0.00 [-0.08]	0.14 [1.74]	0.18 [2.45]	0.18 [2.58]	-0.10 [-2.42]	0.06 [1.01]
Two Factor Model (Market + Trade)							
α (%)	0.00 [0.00]	0.15 [1.18]	0.08 [0.53]	0.01 [0.08]	-0.14 [-0.72]	-0.12 [-0.66]	-0.09 [-0.60]
β_{MKT}	-0.01 [-0.15]	-0.04 [-1.04]	0.05 [1.02]	0.11 [2.04]	0.12 [2.24]	-0.11 [-2.86]	-0.01 [-0.26]
β^T	0.69 [6.38]	0.37 [3.15]	0.94 [9.85]	0.72 [5.51]	0.71 [5.96]	0.17 [2.19]	0.78 [7.72]

Alternative measures and spanning by the CC-based trade factor.

5.8 Tables 5 and 6: Country concentration interacted with China exposure

Read: Table 5 double-sorts firms by country concentration and China import ratio. The high-China, high-CC group earns especially high returns. Table 6 shows that high-China firms have different trade-network characteristics and larger China ratios.

Economic message. China is a key driver because it is a central, policy-sensitive node in global supply chains. But the priced object is concentrated dependence on China, not China exposure in isolation.

Table 5: Pricing Results for Portfolios Sorted by Country Concentration $\times$ China Import Ratio

This table reports the empirical results from a conditional bivariate portfolio-sorting procedure based on firms' industry-demeaned country concentration (CC) and China Import Ratio, defined as the share of a firm's total import value sourced from China in a given shipment year. We first sort all stocks into three CC portfolios (P1, P2, P3). Within each CC portfolio, we compute the median China Import Ratio and classify firms above the median as High China and those below the median as Low China. Panel A reports the value-weighted excess returns and the average CC level for each double-sorted portfolio. It also reports several portfolio differences: (i) the difference in returns between P3 and P1 from the High-China and Low-China groups; (ii) the difference between High and Low China within each CC portfolio; and (iii) the difference-in-differences measure, defined as $(P3 - P1)$ in High China minus minus $(P3 - P1)$ in Low China. Panel B reports risk-adjusted returns (alphas) from the Fama-French five-factor model for each double-sorted value-weighted portfolio, along with the corresponding factor loadings (betas). The sample covers January 2014 to June 2025. Newey-West (1987) t -statistics are reported in square brackets.

Panel A: Excess Returns and Country Concentration (Industry Demeaned)

Excess Ret (%)	P1	P2	P3	P3-P1	CC	P1	P2	P3	P3-P1
High China	0.84 [2.99]	1.28 [2.81]	2.03 [4.41]	1.18 [3.30]	High China	-0.30 [-112.23]	-0.06 [-18.96]	0.27 [75.39]	0.57 [104.58]
Low China	0.80 [3.01]	0.74 [2.29]	0.93 [3.44]	0.13 [0.56]	Low China	-0.31 [-156.94]	-0.05 [-13.71]	0.29 [124.72]	0.60 [252.07]
High-Low China	0.04 [0.31]	0.54 [1.47]	1.10 [2.46]	1.06 [2.30]	High-Low China	0.01 [2.02]	-0.01 [-9.97]	-0.03 [-6.92]	-0.03 [-5.06]

Panel B: Fama-French Five-Factor Pricing Results

α (%)	P1	P2	P3	P3-P1	β^{MKT}	P1	P2	P3	P3-P1
High China	-0.13 [-0.94]	0.05 [0.28]	0.68 [2.91]	0.81 [2.59]	High China	0.99 [25.57]	1.18 [18.07]	1.17 [18.50]	0.18 [2.14]
Low China	-0.09 [-0.81]	-0.09 [-0.42]	0.18 [0.83]	0.27 [1.07]	Low China	0.85 [29.02]	0.86 [25.39]	0.84 [16.17]	-0.01 [-0.20]
High-Low China	-0.05 [-0.36]	0.14 [0.52]	0.50 [1.50]	0.55 [1.41]	High-Low China	0.14 [3.71]	0.32 [3.99]	0.33 [3.12]	0.19 [1.64]
β^{SMB}	P1	P2	P3	P3-P1	β^{HML}	P1	P2	P3	P3-P1
High China	0.05 [0.77]	-0.15 [-1.45]	-0.14 [-1.24]	-0.19 [-1.17]	High China	0.10 [1.42]	-0.41 [-4.41]	-0.35 [-2.91]	-0.45 [-2.56]
Low China	-0.06 [-1.43]	0.05 [0.77]	0.27 [3.62]	0.33 [3.90]	Low China	0.10 [2.00]	0.08 [1.11]	0.13 [1.19]	0.03 [0.25]
High-Low China	0.11 [1.44]	-0.19 [-1.57]	-0.41 [-2.60]	-0.52 [-2.50]	High-Low China	0.00 [-0.08]	-0.49 [-3.79]	-0.48 [-2.47]	-0.48 [-2.19]
β^{RMW}	P1	P2	P3	P3-P1	β^{CMA}	P1	P2	P3	P3-P1
High China	0.19 [1.94]	-0.08 [-0.58]	0.34 [2.47]	0.15 [0.72]	High China	0.12 [1.29]	0.01 [0.07]	-0.18 [-0.87]	-0.30 [-1.13]
Low China	0.24 [3.49]	0.06 [0.95]	0.09 [0.95]	-0.15 [-1.29]	Low China	0.15 [1.57]	0.02 [0.12]	0.04 [0.28]	-0.12 [-0.78]
High-Low China	-0.05 [-0.59]	-0.14 [-0.84]	0.25 [1.30]	0.30 [1.25]	High-Low China	-0.03 [-0.59]	-0.01 [-0.03]	-0.21 [-0.85]	-0.18 [-0.71]

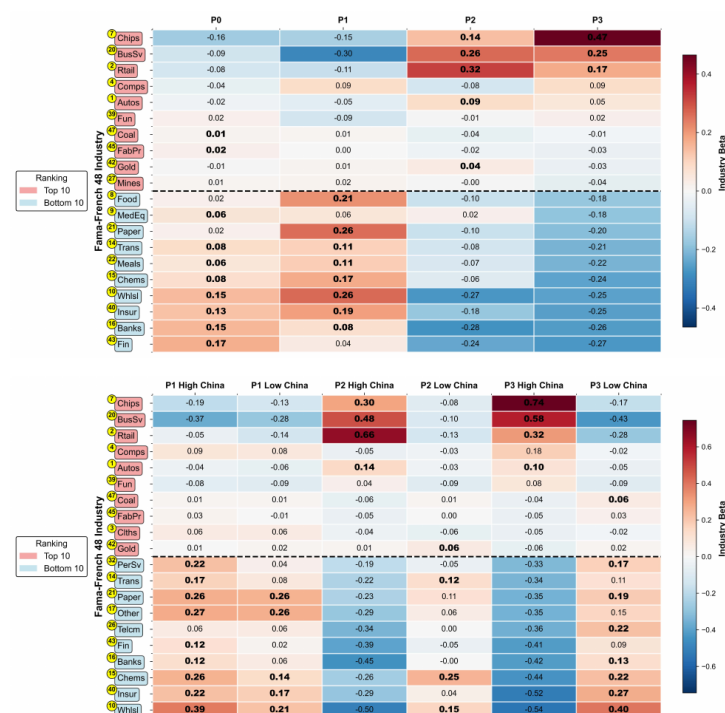
Pricing results for country concentration interacted with China import ratio.

5.9 Figure 5 and Figure 6: Industry decomposition

Read: Figure 5 shows industry exposures of the CC portfolios and China double-sort portfolios. Figure 6 decomposes trade metrics and pricing premia across industries.

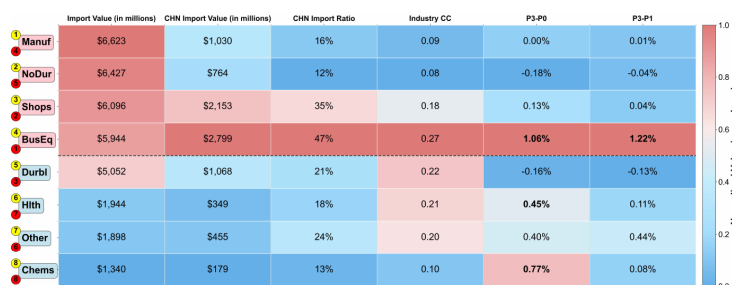
Economic message. Business Equipment and related technology-intensive sectors contribute strongly to the premium, consistent with the idea that modern high-tech production depends on complex and geographically concentrated supply chains.

Figure 5: Industry Beat Exposure of CC-Portfolios and CC $\times$ High vs. Low China Portfolios



Industry beta exposure of trade-vulnerability portfolios.

Figure 6: Industry Level Trade and Risk Premium Analysis



Industry-level trade metrics and trade-risk premia.

5.10 Table 7: Industry-controlled portfolios

Read: The premium survives both global industry-neutral weighting and within-industry sorting. The industry-neutral HL30 alpha is smaller than the baseline but remains significant. The within-industry HL30 premium is also significant.

Economic message. Industry composition is part of the economic story, but not the whole story. Concentrated sourcing is priced even after comparing firms within similar industries.

Table 7: Industry-Controlled CC Portfolios

This table reports the pricing results for trade-vulnerability (CC) portfolios constructed using two industry-neutralization approaches: (i) global industry-neutral weighting and (ii) within-industry sorting. Under the global industry-neutral method, we first form CC portfolios based on Fama–French 48 industry-demeaned CC. For each CC portfolio, we compute value-weighted returns for every industry represented in the portfolio and then reweight these industry-level returns using global market-value weights computed from the full sample of Fama–French 48 industries. Because CC portfolios may not span all 48 industries, the long–short factors (HL30 and HL31) are constructed only from the intersection of industries appearing in both legs of the long–short portfolios (P0–P3 or P1–P3) to ensure exact industry neutrality. Under the within-industry sorting method, firms are sorted into CC portfolios separately within each Fama–French 12 industry. We exclude industries with fewer than 20 firms on average (Energy, Telecom, Utilities, and Money). The final CC portfolios are obtained by aggregating the eight remaining industries using global industry market-value weights. Panel A and Panel B report excess returns as well as CAPM and Fama–French five-factor alphas for the global industry-neutral portfolios and the within-industry portfolios, respectively. The sample spans January 2014 to June 2025. Newey–West (1987) t -statistics are reported in brackets.

Panel A: Global Industry Neutral Weighting						
	P0	P1	P2	P3	HL30	HL31
Excess Ret(%)	0.86 [2.74]	0.86 [2.85]	1.04 [3.09]	1.26 [4.02]	0.42 [3.96]	0.41 [3.19]
$\alpha^{CAPM}(\%)$	-0.15 [-2.43]	-0.10 [-1.05]	0.05 [0.39]	0.28 [2.99]	0.46 [4.16]	0.39 [3.42]
$\alpha^{FF5}(\%)$	-0.04 [-0.85]	-0.06 [-0.76]	0.15 [1.25]	0.31 [3.79]	0.38 [3.98]	0.38 [3.12]

Panel B: Within Industry Sorting						
	P0	P1	P2	P3	HL30	HL31
Excess Ret(%)	1.03 [2.97]	1.03 [3.40]	1.00 [3.31]	1.70 [5.01]	0.67 [3.93]	0.67 [3.07]
$\alpha^{CAPM}(\%)$	-0.01 [-0.06]	0.13 [1.13]	0.05 [0.39]	0.68 [4.72]	0.69 [4.16]	0.56 [2.79]
$\alpha^{FF5}(\%)$	0.07 [0.80]	0.08 [0.75]	-0.01 [-0.10]	0.63 [4.36]	0.56 [3.51]	0.55 [2.71]

Industry-neutral and within-industry pricing results.

5.11 Tables 8 and 9: Technology exposure and pricing technology portfolios

Read: Table 8 shows that technology and semiconductor factors do not eliminate the trade-factor alpha. Table 9 shows the reverse: adding the trade factor to CAPM largely removes the abnormal performance of technology and semiconductor portfolios, and those portfolios load strongly on R^{Trade} .

Economic message. The trade factor helps reinterpret part of technology and semiconductor outperformance as compensation for bearing global supply-chain fragility.

Table 8: Tech Exposure of Trade Vulnerability Portfolios

This table reports the technology-exposure regressions for the global industry-neutral trade-vulnerability portfolios and factor. For each value-weighted portfolio (P0, P1, P2, P3) and for the high-minus-zero factor (HL30), we estimate technology betas using equation (6). In the left panel, we use three broad technology indices as the technology factor: the Nasdaq Composite (XCMP), the MSCI USA Information Technology Index (MXUS0IT), and the S&P 500 Information Technology Index (S5INFT). In the right panel, we use three semiconductor-specific indices as alternative technology proxies: the MSCI ACWI Semiconductors & Semiconductor Equipment Index (MXWD0ST), the S&P 500 Semiconductors & Semiconductor Equipment Industry Index (S5SSEQX), and the Fama-French Chips industry portfolio. For each specification, we report the risk-adjusted return (α), the market beta, and the technology-factor beta (β^{TC}), where the factor corresponds to the index listed in the column header. The sample spans January 2014 to June 2025. Newey–West (1987) t -statistics are reported in brackets.

	Technology Index							Semiconductor Index					
	P0	P1	P2	P3	HL30	HL31		P0	P1	P2	P3	HL30	HL31
$\alpha(\%)$	-0.12	-0.01	0.07	0.28	0.40	0.29	$\alpha(\%)$	-0.12	-0.06	0.05	0.26	0.37	0.32
	[-1.92]	[-0.18]	[0.54]	[3.05]	[3.49]	[2.47]		[-2.09]	[-0.64]	[0.43]	[2.75]	[3.77]	[3.02]
β^{MKT}	1.04	1.00	1.02	1.01	-0.03	0.01	β^{MKT}	1.04	1.00	1.02	1.01	-0.03	0.01
	[66.06]	[30.46]	[44.59]	[30.44]	[-0.99]	[0.24]		[66.86]	[30.58]	[42.62]	[29.04]	[-1.00]	[0.25]
XCMP	-0.18	-0.54	-0.11	-0.02	0.17	0.52	MXWD0ST	-0.05	-0.06	0.00	0.03	0.09	0.09
	[-3.10]	[-7.29]	[-1.18]	[-0.27]	[1.81]	[4.84]		[-2.39]	[-2.19]	[-0.11]	[1.70]	[3.28]	[2.77]
$\alpha(\%)$	-0.06	0.02	0.14	0.28	0.33	0.26	$\alpha(\%)$	-0.10	-0.06	0.06	0.25	0.35	0.31
	[-1.07]	[0.22]	[1.07]	[2.93]	[3.00]	[2.33]		[-1.90]	[-0.56]	[0.52]	[2.62]	[3.57]	[2.81]
β^{MKT}	1.04	1.00	1.02	1.01	-0.03	0.01	β^{MKT}	1.04	1.00	1.02	1.01	-0.03	0.01
	[83.12]	[29.34]	[49.27]	[29.95]	[-0.96]	[0.24]		[67.07]	[30.70]	[43.58]	[28.84]	[-0.99]	[0.25]
MXUS0IT	-0.20	-0.27	-0.19	0.01	0.21	0.28	S5SSEQX	-0.05	-0.05	-0.01	0.03	0.09	0.09
	[-6.23]	[-4.91]	[-3.15]	[0.35]	[5.11]	[4.17]		[-2.88]	[-2.09]	[-0.45]	[1.87]	[3.58]	[2.82]
$\alpha(\%)$	-0.04	0.02	0.14	0.27	0.32	0.25	$\alpha(\%)$	-0.05	0.00	0.12	0.24	0.29	0.24
	[-0.85]	[0.27]	[1.11]	[2.93]	[2.93]	[2.28]		[-0.96]	[0.00]	[0.94]	[2.46]	[2.80]	[2.24]
β^{MKT}	1.04	1.00	1.02	1.01	-0.03	0.01	β^{MKT}	1.04	1.00	1.02	1.01	-0.03	0.01
	[86.67]	[29.27]	[49.14]	[29.94]	[-0.96]	[0.24]		[77.90]	[30.64]	[45.03]	[28.88]	[-0.98]	[0.24]
S5INFT	-0.21	-0.25	-0.18	0.01	0.22	0.26	Chips	-0.12	-0.13	-0.09	0.05	0.18	0.18
	[-7.03]	[-5.00]	[-2.99]	[0.37]	[5.37]	[4.15]		[-5.17]	[-3.97]	[-2.59]	[2.52]	[6.39]	[4.70]

Technology exposure of trade-vulnerability portfolios.

Table 9: Pricing the Tech Performance using the Trade Risk Factor

This table reports the pricing results for the global industry-neutral trade-vulnerability factor applied to technology portfolios. For each technology index, we estimate the two-factor model in equation (7). We employ three broad technology indices as technology-portfolio proxies—the Nasdaq Composite (XCMP), the MSCI USA Information Technology Index (MXUS0IT), and the S&P 500 Information Technology Index (S5INFT). As alternative proxies, we also consider three semiconductor indices: the MSCI ACWI Semiconductors & Semiconductor Equipment Index (MXWD0ST), the S&P 500 Semiconductors & Semiconductor Equipment Industry Index (S5SSEQX), and the Fama-French Chips industry portfolio. As a benchmark, we additionally report CAPM estimates for each technology portfolio. The left panel presents results based on the broad technology indices, and the right panel presents results based on the semiconductor indices. The upper panel reports the average excess returns of each technology portfolio. The middle panel reports the CAPM estimates. The lower panel reports the results from the two-factor model in equation (7). For each specification, we report the risk-adjusted return (α), the market beta, and the trade-factor beta (β^T). The sample spans January 2014 to June 2025. Newey–West (1987) t -statistics are shown in brackets.

	Technology Index				Semiconductor Index		
	XCMP	MXUS0IT	S5INFT		MXWD0ST	S5SSEQX	Chips
Excess Ret(%)	1.23	1.53	1.57		1.89	2.18	1.97
	[3.32]	[3.98]	[4.18]		[3.13]	[3.55]	[4.12]
CAPM				CAPM			
$\alpha(\%)$	0.17	0.45	0.51	$\alpha(\%)$	0.66	0.91	0.8
	[1.06]	[2.28]	[2.52]		[1.81]	[2.24]	[2.76]
β^{MKT}	1.1	1.11	1.1	β^{MKT}	1.27	1.32	1.22
	[30.53]	[22.15]	[20.59]		[10.74]	[9.84]	[16.77]
Two Factor Model (Market + Trade)				Two Factor Model (Market + Trade)			
$\alpha(\%)$	0.08	0.18	0.21	$\alpha(\%)$	0.3	0.46	0.27
	[0.46]	[0.85]	[0.96]		[0.85]	[1.26]	[0.98]
β^{MKT}	1.1	1.13	1.12	β^{MKT}	1.3	1.35	1.25
	[29.24]	[20.96]	[19.64]		[10.76]	[9.84]	[16.40]
β^T	0.21	0.63	0.7	β^T	0.84	1.05	1.21
	[1.78]	[5.82]	[6.15]		[3.45]	[3.45]	[6.36]

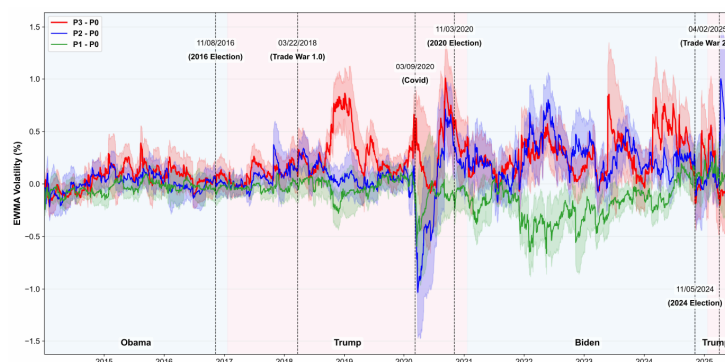
Pricing technology and semiconductor performance using the trade factor.

5.12 Figures 7 and 8: Time-varying risk and tariff-event evidence

Read: Figure 7 shows that volatility spreads between vulnerable portfolios and the zero-trade benchmark rise around major trade-risk episodes. Figure 8 shows cumulative abnormal returns around the 2025 tariff announcements.

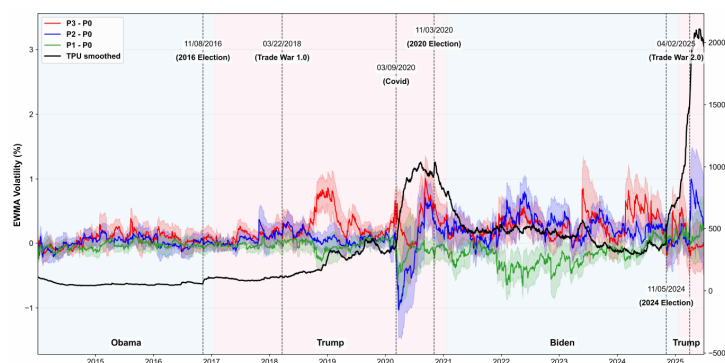
Economic message. Trade risk is dynamic. The vulnerable set changes with the trade-policy regime: broad tariff shocks hurt medium-vulnerability firms, while China-specific escalation hits the concentrated China-dependent firms more directly.

Figure 7(a): Volatility of CC-Portfolios (P1, P2, P3) against The Zero Trade Exposure Portfolio (P0)



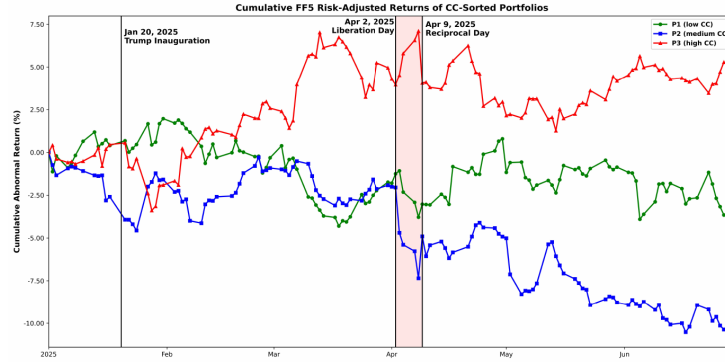
Volatility spreads of CC portfolios relative to zero-trade firms.

Figure 7(b): Volatility of CC-Portfolios (P1, P2, P3) against The Zero Trade Exposure Portfolio (P0) vs. Trade Policy Uncertainty Index



Volatility spreads compared with trade-policy uncertainty.

Figure 8: Performance of CC Portfolios on Liberation Day



Cumulative abnormal returns around the 2025 tariff episodes.

6 Short summary

- The paper builds a firm-level trade-vulnerability measure from bill-of-lading data.
- The main measure is import-country concentration, a Herfindahl index of import shares across source countries.
- Stocks with high country concentration earn higher average returns than stocks with zero trade exposure or diversified trade exposure.
- The high-minus-zero trade factor earns about 0.87% per month and remains significant after standard factor adjustment.
- Alternative trade measures contain some information, but the CC-based factor largely spans them.
- China exposure is central when paired with high concentration; China exposure alone is not the whole story.
- The premium survives industry-neutral and within-industry controls, so it is not just a technology-sector tilt.
- Technology and semiconductor portfolios load on the trade factor, suggesting that some tech outperformance reflects compensation for global supply-chain fragility.
- Volatility and event-study evidence show that the priced risk intensifies around trade-war, COVID, and tariff episodes.

7 Conclusion

Liu and Pan argue that the post-2018 trade-policy environment turns firm-level sourcing concentration into a priced systematic risk. The paper's key innovation is to measure the fragility of a firm's import network directly from bill-of-lading data and then test whether this fragility earns a premium in equity markets.

The central empirical result is strong: firms with highly concentrated import-country exposure earn substantially higher returns than firms with zero or diversified trade exposure, and this premium survives standard asset-pricing controls. The mechanism is also economically coherent. Concentrated import networks expose firms to tariffs, geopolitical conflict, production bottlenecks, and supply-chain disruptions. As these risks become more systematic, investors demand compensation for holding the firms most exposed to them.

The paper's most interesting broader implication is that trade vulnerability connects international trade, production networks, and asset pricing. The same supply-chain structures that make global production efficient in normal times can make firms fragile in bad trade-policy states. That fragility is especially important for technology and semiconductor firms, whose production networks are both global and concentrated. The trade-vulnerability factor therefore provides a market-based way to measure how the financial system prices the resilience and fragility of global value chains.